

Rethinking Human-AI Collaboration in Information Seeking: Why Epistemic Incompatibility Demands New Design Paradigms

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Abstract

The integration of artificial intelligence agents into information retrieval systems has prompted two dominant narratives: AI as replacement for human information seeking, and AI as collaborative partner. This perspective paper challenges both framings by proposing a third alternative – that human information seeking behavior and AI agent functionality represent fundamentally incompatible epistemological paradigms. Drawing on information behavior theory and critical AI scholarship, we argue that this incompatibility stems from three dimensions: epistemic orientation (learning vs. pattern reproduction), temporal structure (processual vs. instantaneous), and agentic purpose (uncertainty resolution vs. task execution). Rather than forcing integration through collaboration metaphors or accepting wholesale replacement, we advocate for system designs that acknowledge and preserve these fundamental differences. This reframing has profound implications for the design of information interaction systems, evaluation methodologies, and research directions in human information interaction and retrieval.

CCS Concepts

• **Information systems** → **Information retrieval**; **Users and interactive retrieval**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

AI agents, information seeking, epistemic incompatibility, human-AI interaction, generative AI, search behavior

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1 Introduction

The rapid deployment of artificial intelligence (AI) agents powered by large language models (LLMs) has fundamentally

altered the landscape of information interaction [48]. Systems such as Microsoft Copilot, Google Gemini, and ChatGPT now mediate billions of information seeking transactions daily, compressing what were once iterative search processes into singular conversational exchanges. This transformation has generated two dominant narratives regarding the future of human information interaction: the *replacement paradigm*, which envisions AI agents supplanting human information seeking activities, and the *collaboration paradigm*, which frames humans and AI as synergistic partners in knowledge work [51].

Both narratives share a critical assumption: that human information seeking behavior and AI agent functionality occupy compatible epistemological spaces that can either substitute for or complement one another. This perspective paper challenges this foundational premise. We argue that human information seeking and AI agent operation represent fundamentally *incompatible* epistemological paradigms; not merely different approaches to the same goal, but categorically distinct modes of engaging with information that serve different purposes, operate through different mechanisms, and produce different kinds of knowledge.

This incompatibility manifests across three critical dimensions as shown in Figure 1. First, at the *epistemic level*, human information seeking is oriented toward resolving genuine uncertainty and constructing mental models [17, 37], while AI agents execute pattern-matching operations that reproduce statistical regularities from training data without experiencing epistemic states [8]. Second, at the *temporal level*, human search unfolds as a processual activity where learning emerges through iteration and exploration [35, 50], whereas AI agents compress information provision into instantaneous response generation that eliminates the learning scaffold. Third, at the *agentic level*, humans engage with information systems to satisfy intrinsic needs for understanding, while AI agents operate as task completion mechanisms without possessing needs, intentions, or epistemic goals.

Recognizing this incompatibility has significant implications for the human information interaction and retrieval (CHIIR) community. Rather than optimizing for seamless integration or accepting wholesale replacement, we should design systems that preserve the distinct affordances of human information seeking while strategically deploying AI capabilities where they genuinely augment rather than undermine human cognitive processes. This reframing challenges current design practices, evaluation methodologies, and research agendas.



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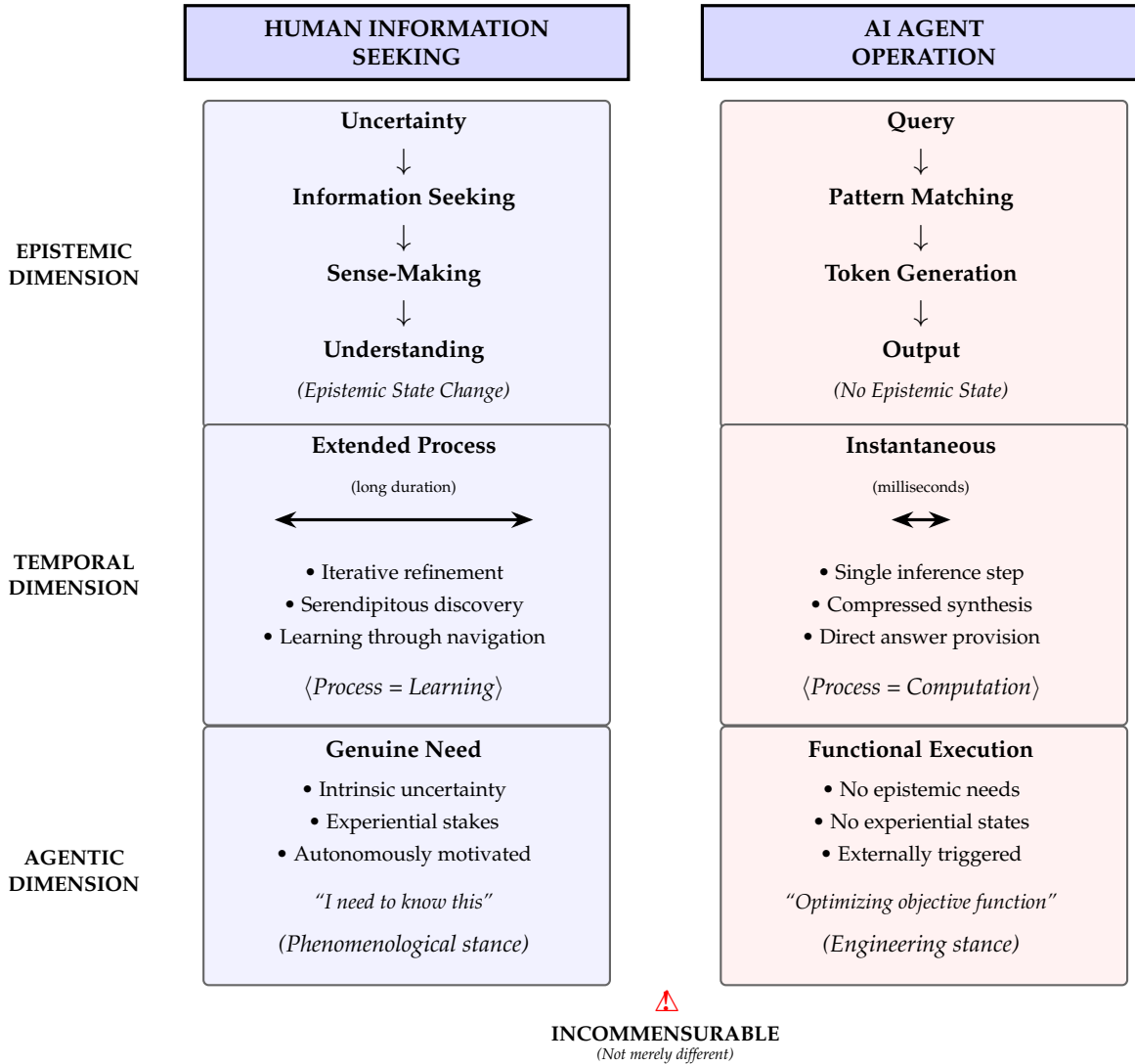


Figure 1: Fundamental Incompatibilities Between Human Information Seeking and AI Agent Operation.

2 Related Work and Theoretical Background

To situate our discussion and arguments about human-AI epistemic incompatibility, we will first review some of the scholarly works as they relate to the information seeking, language models, and human-AI collaboration.

2.1 Information Seeking as Epistemic Activity

Human information seeking has been extensively theorized as a fundamentally cognitive and epistemological endeavor. Information Foraging Theory [37] conceptualizes information seeking as an adaptive process where individuals optimize the ratio of information value to search cost, developing sophisticated strategies analogous to animal foraging behaviors. This framework emphasizes that the *process* of navigation through information spaces serves critical cognitive functions beyond mere answer retrieval.

Dervin's Sense-Making Methodology [16, 17] positions information seeking as a response to gaps in understanding – situations where individuals perceive discontinuities in their mental models and seek information to bridge these gaps. Crucially, sense-making is portrayed not as passive reception but as active construction: individuals interpret, evaluate, and integrate information within existing cognitive frameworks, fundamentally transforming both the information and their understanding [40]. Bates's berrypicking model [6] further illustrates this processual nature, demonstrating that information needs evolve during search as users gather information "bit-at-a-time" across diverse sources, with each discovery potentially redirecting the search trajectory.

Exploratory search research [35, 49, 50] further emphasizes the processual nature of information interaction, distinguishing between lookup tasks (finding known items) and

exploratory activities (learning about domains, investigating relationships, synthesizing knowledge). These exploratory activities are characterized by iterative query reformulation, opportunistic discovery, and gradual refinement of information needs – processes that constitute learning in themselves [3].

2.2 AI Agents and Language Models

Recent advances in LLMs have enabled AI systems to generate fluent, contextually appropriate text across diverse domains [11]. These systems, augmented with retrieval mechanisms through Retrieval-Augmented Generation (RAG) [31], can access external knowledge bases and generate responses grounded in retrieved documents [21]. Contemporary AI agents leverage these capabilities to perform complex, multi-step tasks including information synthesis and reasoning [48].

However, critical scholarship has questioned the nature of these capabilities. Bender and colleagues [8] argue that LLMs function as "stochastic parrots" – systems that reproduce statistical patterns from training data without genuine semantic understanding. The models generate outputs based on probability distributions over token sequences, lacking the intentionality, meaning-making, and epistemic states that characterize human language use [9]. This critique extends to questioning whether systems can truly "understand" information or merely perform sophisticated pattern matching [36].

2.3 Human-AI Collaboration Paradigms

The dominant response to AI capabilities in information systems has been to frame human-AI interaction as collaboration. Research in computer-supported cooperative work (CSCW) explores how humans and AI agents might productively work together [2, 43]. Studies examine trust calibration [52], appropriate reliance [5], and complementary strengths in hybrid intelligence systems [15].

However, this collaboration framework rests on assumptions of compatible goals and commensurable contributions. Recent CSCW discourse questions whether single-user AI interactions constitute genuine "collaboration" [19], suggesting fundamental differences between human-human cooperation and human-AI interaction. Evaluation frameworks for human-AI collaboration [30] struggle to capture the asymmetries in agency, intentionality, and epistemic orientation that characterize these interactions.

3 The Epistemic Incompatibility Framework

We propose that human information seeking and AI agent operation exhibit fundamental incompatibilities across three interconnected dimensions: epistemic, temporal, and agentic. These incompatibilities are not merely design challenges to overcome through better interfaces, but constitute deep structural misalignments between how humans and AI systems relate to information.

3.1 Epistemic Dimension: Learning vs. Pattern Reproduction

Human information seeking is fundamentally oriented toward epistemic state changes – transformations in knowledge, understanding, and cognitive models [17, 37]. When individuals engage in information seeking, they experience genuine

uncertainty, formulate questions that reflect gaps in understanding, and integrate new information into existing mental frameworks through active sense-making [40]. This process involves not merely acquiring facts but constructing coherent understandings, evaluating source credibility, reconciling contradictions, and developing nuanced perspectives [27].

In contrast, AI agents operate through statistical pattern matching without experiencing epistemic states [8]. LLMs generate outputs by computing probability distributions over token sequences based on training data regularities. They do not possess beliefs to update, uncertainties to resolve, or mental models to refine. When an AI agent produces a response to an information query, it executes a computational process of retrieving relevant context and generating probable continuations – fundamentally different from the cognitive activities of learning and understanding [9].

This epistemic asymmetry manifests in concrete ways. Human learners benefit from encountering multiple perspectives, wrestling with contradictions, and gradually refining their understanding through iterative engagement [35]. AI agents, lacking epistemic needs, cannot experience this kind of learning during information interaction. They do not undergo conceptual change, experience "aha" moments, or develop deeper understanding through exposure to information. The pattern reproduction operation, however sophisticated, remains qualitatively distinct from the epistemic transformation that characterizes human learning [36].

3.2 Temporal Dimension: Process vs. Product

Human information seeking unfolds temporally as an extended process with intrinsic value beyond its outcomes [7, 50]. The trajectory through information spaces – query formulation, result examination, iterative refinement, serendipitous discovery – constitutes a learning experience itself [35]. Research on exploratory search demonstrates that the time spent navigating information environments correlates with learning outcomes, as individuals develop familiarity with domain terminology, discover unexpected connections, and refine their information needs through interaction [3, 49]. The searching as learning research agenda has formalized this connection, proposing that iterative, reflective search sessions facilitate critical and creative learning that extends beyond simple information acquisition [38].

Crucially, information seeking often serves goals beyond task completion. Individuals engage in exploratory browsing to satisfy curiosity, maintain awareness of domain developments, or experience the pleasure of discovery [14]. The "slow search" phenomenon [45] reflects how sustained engagement with information environments supports deeper processing and more durable learning compared to rapid answer retrieval.

AI agents, conversely, compress information provision into instantaneous response generation. When using conversational AI systems, users receive synthesized answers without traversing the information landscape themselves [48]. This temporal compression eliminates the iterative, processual aspects of information seeking that scaffold learning. Users do

not experience the gradual refinement of queries, the evaluation of multiple sources, or the synthesis across documents – activities that contribute to conceptual development [40].

This temporal incompatibility creates a fundamental tension: the very efficiency that makes AI agents valuable for task completion may undermine the learning opportunities embedded in traditional information seeking processes [4]. This tension extends across contexts: when students use AI to instantly generate essay outlines, when researchers employ AI to summarize literature, when clinicians rely on AI for diagnostic support, or when analysts use AI to synthesize market intelligence, they bypass the cognitive work that traditionally supported knowledge construction and professional expertise development [25].

3.3 Agentic Dimension: Intrinsic Need vs. External Function

Human information seeking is driven by intrinsic epistemic needs: genuine requirements for knowledge to resolve uncertainty, make decisions, or satisfy curiosity [17, 27]. These needs emerge from individuals' situated contexts, reflect their personal goals and values, and motivate engagement with information. The searcher has *stakes* in the outcome: finding relevant information matters for achieving purposes that transcend the search activity itself.

AI agents possess no such intrinsic needs. They have no epistemic uncertainties to resolve, no decisions whose outcomes matter to them, no curiosity to satisfy. When an AI agent engages in multi-step reasoning or retrieves information to answer a query, it executes programmed functions without experiencing the query as *its* problem. The system has no preferences over outcomes beyond optimizing against training objectives [8].

This agentic asymmetry complicates the collaboration metaphor. True collaboration presumes aligned or compatible goals where partners contribute toward shared outcomes. But human information seekers and AI agents have fundamentally different relationships to information tasks. The human seeks to resolve genuine uncertainty; the AI executes retrieval and generation operations. The human uses the information to make consequential decisions; the AI has no stake in those consequences [19].

Moreover, the concept of "AI autonomy" in information seeking contexts is misleading. When AI agents autonomously decompose complex queries into sub-tasks, iteratively refine searches, or synthesize information from multiple sources, they exhibit sophisticated behavior without possessing agency in the human sense, *viz.*, operating without needs, intentions, or values [43]. This creates epistemic and ethical complications when AI-mediated information shapes human beliefs and decisions.

4 Implications for Design and Practice

Recognizing epistemic incompatibility between human information seeking and AI agent operation requires fundamental shifts in how we design, evaluate, and deploy information systems. Rather than pursuing seamless integration or accepting wholesale replacement, we must develop approaches that

preserve human epistemic agency while strategically leveraging AI capabilities. Here we present our perspectives on how to go about doing this.

4.1 Rethinking Evaluation Beyond Efficiency

Traditional IR evaluation prioritizes outcome efficiency through metrics like precision, recall, and time-to-answer [41]. Emerging frameworks for generative IR evaluation [22, 46] consider answer quality and factual accuracy. However, these paradigms inadequately capture the epistemic and temporal dimensions we have identified.

Evaluation frameworks acknowledging incompatibility must assess both outcome quality and process preservation. Key questions include: Does the system support or suppress user learning? Are opportunities for serendipitous discovery maintained? Does interaction cultivate domain knowledge or foster dependence on AI intermediation? Consider two systems that both successfully answer a complex information need: one provides an AI-generated summary directly, while another guides users through structured exploration of sources. Traditional metrics might favor the former for efficiency, but the latter better preserves learning opportunities [35].

4.1.1 Process-Oriented Evaluation Metrics. We propose evaluation metrics complementing traditional measures (Table 1). This is based on a strong belief that research in this direction must develop:

Learning outcome measures that assess whether users develop comparable domain knowledge across AI-mediated and traditional interfaces. Longitudinal studies should track knowledge retention, conceptual understanding, and ability to apply learned information across varying time horizons [26, 47].

Epistemic agency metrics that quantify user control over search strategies and exposure to diverse perspectives [35]. Research in health information seeking suggests that AI systems may require reconceptualization of information literacy and behavioral norms, pointing to the need for metrics capturing these shifts [34].

Cognitive effort assessment that distinguishes productive cognitive load enhancing learning from unproductive friction [10, 40]. Contrary to traditional efficiency assumptions, some cognitive effort in information seeking serves valuable purposes. Research should integrate insights from cognitive psychology with information behavior research [40].

4.1.2 Contextual Evaluation Frameworks. Evaluation must account for epistemic context. A system optimized for rapid fact retrieval may suit lookup tasks but prove detrimental for exploratory learning [3]. Recent expert consensus emphasizes the need for new evaluation approaches considering user modeling and personalization in AI-powered information access systems [1]. We extend this by proposing evaluation frameworks that explicitly consider epistemic goals:

- *Lookup tasks:* Efficiency-oriented metrics remain appropriate when epistemic goals are satisfied through rapid answer retrieval.
- *Exploratory search:* Process engagement metrics become primary, with efficiency as secondary consideration [50].

Table 1: Framework for Process-Oriented Evaluation of Information Systems.

Evaluation Dimension	Traditional IR Metrics	Proposed Process-Oriented Metrics
Outcome Quality	Precision, Recall, nDCG, Time-to-Answer	Answer accuracy, Factual consistency, Source diversity
Learning & Understanding	(Not typically measured)	Knowledge gain (immediate & retained), Conceptual depth, Ability to explain reasoning
Epistemic Agency	(Not typically measured)	User control index, Query reformulation depth, Source examination breadth, Serendipity rate
Process Engagement	Clicks, Dwell time	Meaningful interaction depth, Critical evaluation instances, Cross-source synthesis
Cognitive Development	(Not typically measured)	Information literacy growth, Critical thinking skills, Domain vocabulary acquisition
Long-term Effects	Session success rate	Knowledge durability (30/90 days), Transfer to novel problems, Metacognitive awareness

- *Investigative research*: Evaluation must assess support for critical evaluation, source triangulation, and synthesis capabilities.
- *Learning-oriented search*: Knowledge acquisition and retention metrics take precedence over immediate task completion.

4.2 Design Principles for Preserving Epistemic Agency

System designs should acknowledge incompatibility through several key principles that strategically separate concerns while enabling productive human-AI interaction.

4.2.1 Separation of Concerns and Transparency. Rather than forcing integration, clearly separate domains where AI task completion is appropriate from contexts where human information seeking processes should be preserved. This begins with taxonomizing information needs by epistemic purpose [35]: Are users seeking to learn about a domain (exploration), locate specific items (lookup), or make decisions requiring deep engagement (investigation)? For lookup tasks with no learning objectives, AI-mediated answer provision may be appropriate. For exploratory or investigative tasks where process engagement contributes to cognitive development, systems should support traditional information seeking behaviors [50].

Systems must communicate clearly what epistemic work AI agents perform versus what remains for humans. When AI summarizes research literature, users must understand that statistical synthesis differs fundamentally from critical evaluation based on domain expertise [32]. Interfaces might distinguish AI-retrieved facts from AI-generated interpretations, and both from human-constructed understanding. Rather than presenting only final outputs, systems could expose retrieval paths, information sources, and synthesis operations, supporting users in developing mental models of AI behavior while maintaining connection to underlying information landscapes [29].

4.2.2 Hybrid Architectures and Adaptive Mediation. Acknowledging incompatibility suggests hybrid architectures that strategically deploy AI capabilities for specific subtasks while preserving overall human control [43]. For example, AI might handle query expansion and relevance ranking while humans

retain agency over result examination and synthesis. Or systems might use AI to generate overview summaries that orient users before facilitating structured exploration of primary sources. Such architectures recognize that human and AI contributions are incommensurable: rather than asking whether humans or AI perform better at information tasks, we should ask which aspects serve cognitive purposes requiring human engagement, and which represent computational overhead AI can productively handle [15].

For contexts where information seeking serves learning objectives, systems should preserve cognitive scaffolding through context-sensitive policies [48]. This might involve progressive disclosure of AI capabilities: initially supporting traditional search to encourage engagement with primary sources, then offering AI summarization after users demonstrate domain familiarization [39]. Educational technologies might deliberately introduce beneficial friction – modest cognitive challenges promoting deeper processing despite reducing immediate efficiency [10].

4.3 Information Literacy for AI-Mediated Environments

The incompatibility framework highlights urgent questions about information literacy. When AI agents mediate information access, what competencies must humans maintain? If pattern-matching systems handle information retrieval and synthesis, do humans risk atrophy of critical evaluation skills? How do we cultivate epistemic agency when information provision becomes increasingly automated [44]?

Users need three critical capabilities here [33]: (1) *AI literacy for information seekers*: understanding how AI systems retrieve, process, and synthesize information, including awareness of limitations like hallucination and bias amplification; (2) *metacognitive awareness*: recognizing when AI delegation undermines learning objectives and understanding cognitive benefits of engagement [39]; and (3) *epistemic evaluation competencies*: assessing AI synthesis quality, recognizing training data biases, and validating AI-provided information against primary sources [27].

Research must address how AI-mediated information seeking affects cognitive development, particularly for students

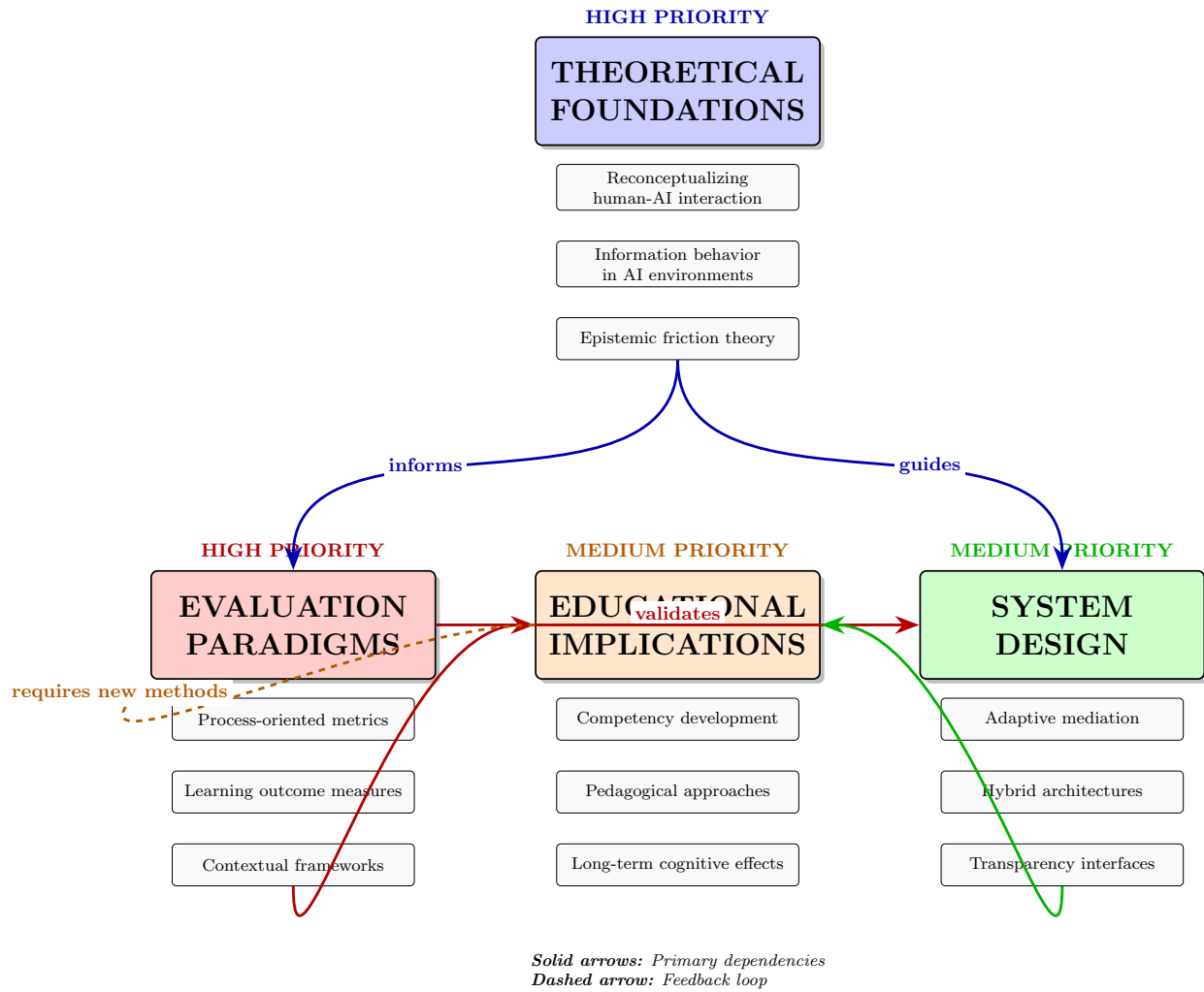


Figure 2: Research agenda for incompatibility-aware information systems. The framework identifies four interconnected research themes arranged in an iterative cycle. Arrows show how each theme informs and depends on others. Theory provides the foundation that informs evaluation approaches and guides system design, while education research requires new evaluation methods, completing the research cycle. Priority levels (high/medium) indicate areas requiring immediate community attention.

and early-career researchers. Studies might investigate whether dependence on AI agents for literature review diminishes critical reading skills, or whether AI assistance enables focus on higher-level synthesis [25]. Educational contexts should introduce AI capabilities progressively, first developing foundational information literacy through traditional search, then teaching appropriate AI usage to prevent premature dependence. Comparative analysis exercises – evaluating AI summaries against original documents – can illuminate how AI synthesis differs from human understanding. Rather than always optimizing for efficiency, educational systems might require students to demonstrate understanding before accessing AI assistance [10].

4.4 Reconceptualizing User Roles and the Search Process

Current frameworks position users as either passive recipients of AI outputs or active collaborators in hybrid intelligence systems. The incompatibility perspective suggests a third conception: users as epistemic agents whose information seeking activities serve intrinsic cognitive purposes that AI cannot fulfill. This reframes design priorities from optimizing human-AI collaboration toward preserving human epistemic agency.

Such preservation requires transparent delineation of when AI assistance genuinely augments human capabilities versus when it substitutes for cognitively valuable human activities.

Systems might explicitly scaffold metacognitive awareness, helping users recognize when AI delegation undermines learning objectives [39]. Design patterns could distinguish between contexts where efficiency justifies AI mediation and contexts where process engagement warrants human control.

Historically, IR research has focused on improving result quality under the assumption that better results serve searchers' needs [41]. The incompatibility framework inverts this emphasis: for many information seeking contexts, particularly exploratory search, the process of seeking carries value independent of results. Eliminating the search process through direct AI answer provision may increase efficiency while undermining intrinsic cognitive benefits [50]. This suggests a fundamental tension in generative IR: the more effectively AI systems answer queries, the more they may undermine the learning scaffolded by traditional search. Research must characterize contexts where this tension emerges versus where AI mediation poses no such conflict [3].

5 A Research Agenda for Incompatibility-Aware Information Systems

Addressing epistemic incompatibility requires comprehensive research across interconnected themes. Figure 2 illustrates our proposed research agenda, organized around four key areas: evaluation paradigms, theoretical foundations, system design, and educational implications. Each area comprises specific research directions that build upon and inform the others, creating an integrated approach to understanding and designing for epistemic incompatibility.

5.1 Theoretical Foundations: Toward Epistemic Interaction Theory

The incompatibility framework challenges existing theoretical constructs in human information interaction (see "Theoretical Foundations" quadrant, Figure 2). Research must develop theoretical foundations that account for the fundamental asymmetries we have identified.

5.1.1 Reconceptualizing Human-AI Interaction. The collaboration metaphor dominating current human-AI interaction research presumes compatible goals and commensurable contributions [2, 43]. Our analysis suggests this framing obscures critical differences. Research should explore alternative frameworks:

Complementarity theory. Rather than collaboration, conceptualize human-AI interaction in terms of complementary but incommensurable capabilities [15]. This framework acknowledges that humans and AI excel at categorically different aspects of information work. Research must characterize these complementarities and identify optimal division of cognitive labor.

Mediation and disintermediation dynamics. Draw on media theory to understand how AI agents mediate access to information in ways qualitatively distinct from prior information technologies [23]. While traditional IR systems mediate access to documents, AI agents mediate access to knowledge itself, raising fundamental questions about epistemic authority and validation.

Epistemic friction theory. Develop theoretical frameworks for productive versus counterproductive friction in information interaction [10]. Some obstacles in traditional search – query reformulation, result evaluation, source comparison – constitute valuable epistemic work. Theory must distinguish beneficial friction from mere inefficiency.

5.1.2 Information Behavior in AI-Mediated Environments. Classical information behavior theories, such as Information Foraging [37], Sense-Making [18], Berrypicking [6], rest on assumptions about user agency and process engagement that AI mediation challenges. Emerging research suggests that AI integration necessitates scrutiny of existing information seeking theory and reconsideration of behavioral norms. Research must extend these theories to account for:

Distributed epistemic agency. When AI agents autonomously decompose queries, retrieve information, and synthesize responses, where does epistemic agency reside? Theory must address how responsibility for knowledge claims distributes across human-AI systems [19].

Sense-making with AI-generated synthesis. Classical sense-making emphasizes active construction of understanding through engagement with information [18]. How does this process transform when AI provides pre-synthesized interpretations? Research should investigate whether users who consume AI-generated summaries develop equivalent conceptual models to users who synthesize information themselves.

Information foraging in AI-augmented environments. Information Foraging Theory posits that users optimize information value per unit cost [37]. The rapid evolution of information retrieval with generative AI technologies is fundamentally altering the cost-benefit calculations in foraging [48]. When AI provides instant answers, does this enhance or undermine the foraging process?

5.2 System Design: Incompatibility-Aware Architectures

Design research must translate theoretical insights into concrete system architectures that preserve human epistemic agency while leveraging AI capabilities (see "System Design" quadrant, Figure 2). Figure 3 illustrates a proposed hybrid architecture embodying these principles.

5.2.1 Adaptive Mediation Frameworks. Systems should dynamically adjust the degree of AI mediation based on epistemic context. Research must develop:

Context-sensitive mediation policies that recognize information seeking contexts (lookup, exploration, learning, investigation) and modulate AI involvement accordingly [48]. For learning-oriented contexts, systems might initially withhold AI synthesis to encourage human engagement, then provide AI assistance after demonstrating baseline understanding.

Progressive disclosure architectures that initially present traditional search interfaces, then progressively introduce AI capabilities as users demonstrate readiness [39]. This approach

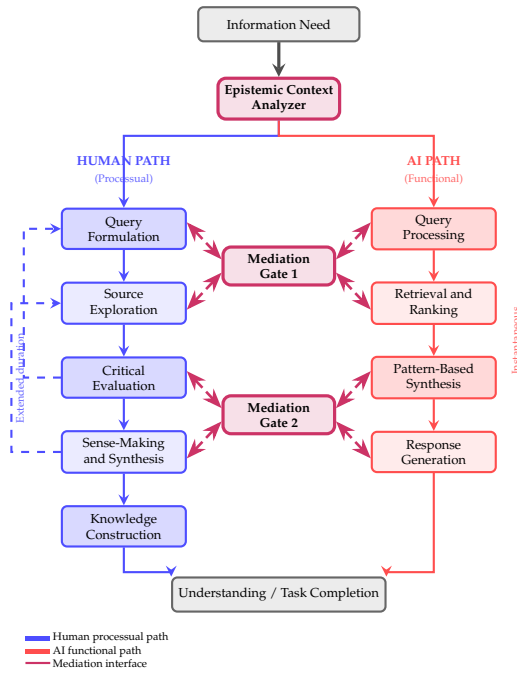


Figure 3: Hybrid human-AI information architecture showing distinct pathways. The human path (left, blue) represents extended, iterative processes supporting learning and sense-making. The AI path (right, red) represents instantaneous functional operations. Mediation gates (middle, purple) serve as controlled interface points where context-aware policies determine when AI outputs inform human processes or when human oversight guides AI operations. Dashed feedback arrows on the human path indicate iterative refinement characteristic of exploratory search.

preserves learning scaffolds while providing efficiency when appropriate.

Metacognitive awareness mechanisms that make AI mediation visible and support user reflection on when AI delegation is appropriate [29]. Systems should prompt users to consider: "Would engaging with primary sources serve your learning goals better than accepting this AI summary?"

5.2.2 Hybrid Human-AI Information Architectures. Rather than AI-mediated or human-driven search, explore hybrid architectures that strategically allocate information work:

Selective AI augmentation. Identify sub-tasks where AI assistance enhances rather than supplants human cognitive processes [24]. For example, AI might handle query expansion and source discovery while humans retain responsibility for source evaluation and synthesis.

AI as epistemic scaffold. Position AI systems as educational scaffolds that support learning rather than bypass it. This might involve AI-generated questions that guide exploration, structured prompts that encourage critical evaluation, or challenges to user assumptions that promote deeper engagement.

Transparency-enhanced AI mediation. Design systems that expose AI reasoning processes, retrieved sources, and synthesis operations [32]. Users should understand not just what information AI provides, but how it was obtained and synthesized, supporting informed evaluation of AI outputs.

5.3 Empirical Investigation and Cross-Cutting Priorities

While we have theoretically articulated epistemic incompatibility, empirical validation remains essential. Research should:

- Compare learning outcomes between AI-mediated and traditional search interfaces across diverse domains and task types.
- Investigate whether users who consume AI-generated syntheses develop different conceptual models than users who synthesize information themselves.
- Examine long-term effects of AI-mediated information seeking on information literacy, critical thinking, and domain expertise development.
- Characterize contexts where AI mediation enhances versus undermines human epistemic goals through controlled experiments and field studies.

5.3.1 Ethical and Societal Considerations. Incompatibility-aware design raises ethical questions requiring sustained attention:

Equity and access. If preserving epistemic agency through traditional search requires greater cognitive effort, does this create inequities? Users with limited time, resources, or education may disproportionately rely on AI mediation, potentially exacerbating epistemic inequality.

Informed consent and transparency. Users should understand when systems mediate their information access through AI and what epistemic tradeoffs this involves. Research must develop transparency mechanisms that support informed choices about AI reliance [20].

Cultural variation. Epistemic values and information seeking norms vary across cultures. Research should investigate how incompatibility manifests differently across cultural contexts and whether design principles require cultural adaptation.

5.3.2 Interdisciplinary Integration. Addressing epistemic incompatibility requires integration across multiple disciplines. Cognitive science informs understanding of learning and memory processes relevant to assessing when AI mediation helps versus hinders cognition. Education provides frameworks for scaffolded learning and beneficial difficulty. Philosophy offers theoretical foundations for understanding knowledge and cognitive states through epistemology and philosophy of mind. Science and Technology Studies (STS) provides critical perspectives on how technologies shape social practices and power relations. Human-Computer Interaction (HCI) contributes design principles and evaluation methodologies. Recent community discussions emphasize the importance of such interdisciplinary approaches in addressing challenges posed by generative AI in information retrieval [48].

5.3.3 Implementation Pathway. Realizing this research agenda requires coordinated effort across the research community. The community should develop shared evaluation frameworks and standardized evaluation protocols for process-oriented metrics, enabling comparison across studies and accumulation of evidence about effective design patterns. Research needs open datasets supporting investigation of incompatibility effects, including longitudinal user studies with learning outcome measures and comparative datasets contrasting AI-mediated and traditional search. Funding agencies should prioritize interdisciplinary research bringing together expertise from information retrieval, cognitive science, education, HCI, and critical AI studies. While academic research can investigate incompatibility effects, industry deployment determines real-world impact. Researchers must engage with industry partners to translate findings into production systems while maintaining independence to critically evaluate AI-mediated designs.

6 Discussion and Conclusion

While the premise of this perspective paper is provocative, the recommendations for the research agenda in this very important area of human-AI "collaboration" warrants careful consideration and extensive discussions by the community.

6.1 Broader Implications and Limitations

The epistemic incompatibility framework challenges prevailing assumptions in AI research beyond information retrieval. Much discourse on AI systems presumes that improving capability, such as making AI more helpful, accurate, and autonomous, universally benefits users [12]. Our analysis suggests this presumption fails for activities where human cognitive engagement carries intrinsic value. In this sense, our framework extends the longstanding interactive IR commitment to process-oriented, human-centered research into the AI era, demonstrating the continued relevance of foundational Interactive IR (IIR) principles even as underlying technologies transform [26].

This has implications across domains. In education, AI tutoring systems that solve problems for students may reduce learning despite improving immediate task performance [39]. In decision support, AI systems that automate judgment may undermine development of expertise [13]. In creative work, AI co-creation tools may enhance productivity while attenuating the cognitive benefits of creative struggle [51].

Our framework identifies incompatibilities but does not prescribe universal solutions. Different contexts require different balances between efficiency and process preservation. Future work must empirically characterize when AI mediation genuinely augments human information interaction versus when it undermines learning and epistemic development. Longitudinal studies should investigate how sustained use of AI-mediated information systems affects cognitive capabilities, information literacy, and epistemic agency. Do users develop productive complementarity with AI tools, or does dependence on AI mediation atrophy critical thinking skills? How do different system designs shape these developmental trajectories?

Additionally, our analysis focuses on individual information seeking. Collaborative information seeking and organizational knowledge work present further complexities [42]. How does AI mediation affect collective sense-making processes? Do AI agents facilitate or impede productive disagreement and perspective-taking in group contexts?

6.2 A Call for Thoughtful Deployment

The IR and HCI communities stand at a critical juncture. As AI agents become increasingly sophisticated and ubiquitous, we must decide whether to embrace wholesale transformation of information interaction or to advocate for preserving cognitive affordances that have characterized human information seeking. This decision carries consequences for human intellectual life.

We argue for a third path: designing systems that acknowledge fundamental incompatibilities while strategically leveraging AI capabilities. This requires resisting technological determinism – the belief that AI advancement necessitates adoption across all contexts – in favor of value-sensitive design that prioritizes human flourishing [20].

This perspective paper challenges the dominant narratives framing AI agents as either replacements for or collaborators with human information seekers. We propose that human information seeking and AI agent operation exhibit fundamental incompatibilities across three interconnected dimensions: epistemic, temporal, and agentic. We use "incompatibility" rather than strict Kuhnian "incommensurability" to emphasize that while these modes of information engagement can be compared, they lack common measures for direct evaluation – a distinction Kuhn himself later clarified [28]. These incompatibilities are not merely design challenges to overcome through better interfaces, but constitute deep structural misalignments between how humans and AI systems relate to information.

Rather than seamlessly integrating AI agents into all information seeking contexts, we should preserve domains where human cognitive engagement carries intrinsic value while strategically leveraging AI for specific subtasks. Evaluation methodologies must assess process preservation alongside outcome quality. Information literacy education must cultivate metacognitive awareness about when AI delegation undermines learning. System designs should separate concerns, maintaining human epistemic agency while benefiting from AI capabilities.

As the IR, HCI, and other communities navigate an AI-transformed landscape, we must advocate for human information seeking as a valuable cognitive activity – not merely a means to acquire information but a process through which understanding emerges. The future of information interaction should not be determined solely by technological capabilities but guided by careful consideration of what makes human engagement with information meaningful, educational, and epistemically valuable.

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